BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 4 Semester: Spring26
Duration: 25minutes Full Marks: 15

CSE 340: Computer Architecture

Name: Solution	ID:	Section:

1. Assume that a memory access takes 20 ns if it results in a cache hit, and 50 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below:

Simulation

Index	Valid	Tag	Data
000	1	~~100~~ ~~111~~ 010	~~Data[100000]~~ ~~Data[111000]~~ Data[010000]
001	0		
010	~~0~~ 1	011	Data[011010]
011	0		
100	1	100	Data[100100] ✓
101	~~0~~ 1	~~011~~ 100	~~Data[011101]~~ Data[100101]
110	1	011	Data[011110] ✓
111	1	111	Data[111111] ✓

The size of a block is 8 bits. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below. [8]

block size = 8 bits = 1 byte → no offset

Mem add.
MSB XXX XXX LSB
Tag → 3 bits
index → 3 bits

Memory/Address	Hit/Miss	Access Time
36 = 100100	Hit	20
56 = 111000	Miss	50
26 = 011010	Miss	50
63 = 111111	Hit	20
30 = 011110	Hit	20

16 = 010000	Miss	50
29 = 011101	Miss	50
37 = 100101	Miss	50
Total Access Time		310 ns

a) Complete the cache table simulation in the given table above. [2]

b) Find the total access time. [1] 310 ns

6 bytes → 0 to 5 → 101 → 3 bits ← offset size

Q2: Suppose a cache has 16 blocks. Each block can store 48 bits of data.

a) What is the size of index bits? [1]
b) Using the mentioned cache system, for a memory address $\underbrace{101}_{\text{Tag}}\underbrace{1100}_{\text{index}}\underbrace{010}_{\text{offset}}$, Find the index, tag and offset bits. [3]

Answer:

a) 4 bits.

b) Index bits = 1100

Offset bits = 010

Tag bits = 101